Tactix: PB versus DSR on a Winner Family

Report to Andrew

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Outcome

On the same archived `or_randxor` UNSAT input, LymphoSAT reproduced its 157,075-byte verified PBP proof while Tactix produced a 7,770,045-byte native DSR proof. Both certificates independently verify. This is a concrete proof-representation/technique gap, not missing semantic UNSAT coverage and not an N2 failure. AM410 separately showed Tactix's startup PBP already covers the exact Tseitin winner witness; the general post-startup PB action interface remains absent.

Evidence and failed first check

Both solvers ran sequentially in hwulab job `1721443` with four CPUs and 32 GiB. LymphoSAT selected `or_randxor`, emitted the exact baseline task-414 PBP bytes, and passed pinned VeriPB and CakePB. Tactix source `6c176683` exited UNSAT after 121,749 conflicts in 4.45 seconds, with zero startup parity recognition. Its first proof-check attempt was never run: the diagnostic script omitted the checker work directory, so the job ended `FAILED 1:0` on a file-creation error. The omission is fixed in the probe runner, not in production code.

Checker-only job `1721451` used the unchanged exact CNF and DSR proof, completed `0:0`, and passed frozen no-option DSR-to-LSR elaboration with independent Trestle checking in both backward and forward every-lemma audits. The AM411 technical report and compressed evidence under `benchmarks/` retain all commands, full original proofs, checks and hashes. A redundant 33 MB archive of reproducible derived LSRs was removed; the raw local files and lean 3.1 MB evidence archive remain.

Interpretation and next actions

The same-allocation times were 0.02 seconds for LymphoSAT and 4.45 seconds for Tactix, but one sequential row is not a competitiveness result. The PB certificate is about 49.5 times smaller. A general typed PB derivation and format/state contract could address such gaps; a family-name dispatcher or ad hoc PBP/DSR splice would violate the shared-rule architecture. Finish the active Opus fuzz and full Debug verdicts, repair the remaining H2 and post-finalization API bugs, then rerun combined N1–N4 and Rank/B+C/WV gates. The requested full Tier-D 5,000-second run remains deferred until those conditions are met. Production online learning still needs the shared default-exact slot controller and safe-point owned sub-strategy switching.